Tetrad-Based Departure Decomposition for FLRW Anisotropy

Final Results Report — Credible-Claim Distillation

BASS / HTT / MIO / OBSSTAT stack
Soongsil OMEG Institute

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Abstract

This report distills the *credible, defensible* results of the tetrad-based departure-decomposition program into a single self-contained document. It deliberately excludes development history, superseded paths, and any result that does not survive the project's claim-firewall. Every result below is one of four kinds: (i) framework statements about the master departure comparator $x_C = \Sigma^2_{\rm std} - W^2_{\rm std} + \Omega_{\rm tilt} + \Omega_{k,\rm aniso}$ and its claim-tiered diagnostic variables; (ii) conditional analytic theorems—both the EGS-type low-ℓ identities and a multicomponent identifiability and restricted Bianchi-I dynamics suite—each proven symbolically with the Wolfram Engine under explicitly stated hypotheses; (iii) model-independent measurements on real Planck and Cosmicflows-4 data, each calibrated against an isotropic null, a forward mock, or a resampling baseline; and (iv) explicitly labelled transfer-conditional diagnostics. The analytic suites are hardened by an external adversarial audit: every theorem is restated under its exact hypotheses (closure-conditional identities, an estimator sampling-variance statement rather than a universal floor, and an additive depth-transport contrast), the restricted Bianchi-I dynamics are re-derived by an *independent* chain-rule verifier (1000-state conservation and Gauss/Codazzi transport, DOP853/Radau cross-checks), and the estimator mechanics (hierarchical bulk-flow coverage, a curl-suppression non-identifiability no-go, and a look-elsewhere max-scan calibration) are validated on synthetic catalogues with known truth. No result here identifies a Bianchi family, detects an anisotropic geometry, claims native low-ℓ Boltzmann-solver validation, or promotes any diagnostic into a posterior-odds or model-ranking statement. The honest publishable envelope is a methods-plus-bounds framework that recovers the established low-ℓ CMB anomalies as null-calibrated model-independent features, maps the CF4 bulk-flow apex profile, measures a forward-mock-calibrated CF4 bulk flow and its affine velocity-gradient decomposition, and proves an exact dust-FLRW constraint-transport oracle (machine-checked with the Wolfram Engine and an independent xAct canonicalization)—while producing no false anisotropy claim.

Contents

1 Scope and claim discipline

This document is a *final-results distillation*, not a record of how the results were obtained. The full multi-chapter VER06 manuscript (which retains derivations, audit trail, and conditioned legacy appendices) is preserved separately and unchanged. Here we carry only what is currently defensible.

Claim tiers. All current public results sit in a claim-tiered observatory. We use the project's tiers: **C1** framework / disclosure, **C2** transfer-conditional, **C3** observer-side diagnostic, **C4** local/global discrimination candidate (gated). Every result in this report is *diagnostic-only* or *transfer-conditional*; none is a production-validated or geometry-level claim.

What this report does *not* claim. The following remain blocked and are stated here once, up front, so that no later sentence can be read as lifting them:

- no identified Bianchi family and no detected anisotropic geometry;
- no native low-ℓ Bianchi Boltzmann-solver output and no native morphology atlas (both absent in this repository state);
- no promotion of any scalar diagnostic (x_C, Q, Π, F, G_F), direction coherence, or low-ℓ feature into Bianchi-family or geometry evidence;
- no posterior odds, Bayes-factor headline, or model ranking; MIO observatory certificates are diagnostic-only and are kept separate from HTT-owned inference;
- no native-validation label on any external/proxy transfer output.

These blocks are lifted only by future work supplying a native low-ℓ morphology atlas with matched masks, matched randoms, full covariance, calibrated nulls, predictive checks, and family-equivalence gates. None of that is present here, and nothing below depends on it.

2 The tetrad departure framework

Master departure comparator. The program organises FLRW departure through a single signed comparator coordinate,

$$x_C \;=\; \Sigma^2_{\text{std}} \;-\; W^2_{\text{std}} \;+\; \Omega_{\text{tilt}} \;+\; \Omega_{k,\text{aniso}}, \tag{1}$$

where $\Sigma^2_{\rm std}$ is a standardised shear-squared contribution, $W^2_{\rm std}$ a standardised vorticity-squared contribution, $\Omega_{\rm tilt}$ a tilt (peculiar-rapidity) term, and $\Omega_{k,\rm aniso}$ an anisotropic spatial curvature term, each referred to a stated frame (normal, matter, CMB, local-boost, or geometry frame).

Definition 1 (Comparator, not magnitude)**.** x_C in (1) is a *signed* comparator coordinate built to compare a model against the FLRW point, not an invariant anisotropy magnitude. Its sign structure (note the $-W^2_{\rm std}$) means x_C may cancel or change sign; it must never be read as a nonnegative "total anisotropy."

Claim-tiered diagnostic variables. On top of (1) the observatory exposes four observer-side scalars, all diagnostic-only:

- *occupancy* Q, a policy-normalised departure score;
- *exceedance* Π, an empirical exceedance curve of a named score above stated thresholds;
- *filling* $F = x_C/x_{\max}$, the fraction of an algebraic admissible ceiling $x_{\max}$ occupied by the comparator;
- *depth gap* $G_F(z) = F(z)/F(z_{\rm ref})$, the redshift-resolved ratio of the filling.

The algebraic ceiling used throughout is the S2a value $x_{\max} = 9.25 \times 10^{-6}$, the combined shear+tilt+curvature admissible ceiling for the full comparator x_C; it is distinct from, and larger than, the VER06 manuscript's shear-only bound $\Sigma^2_{\rm std} < 6.4 \times 10^{-6}$ (the latter bounds $\Sigma^2_{\rm std}$ alone, the former the whole signed comparator). These scalars summarise a comparator; on their own they do not classify a geometry or a Bianchi family.

Kinematic-bound linkage. The comparator is tied to the Maartens–Ellis–Stoeger (MES) kinematic-bound hierarchy: bounds on CMB multipole anisotropy bound the covariant shear, vorticity, and acceleration, which in turn bound the contributions in (1). We use MES as a derived bound under its stated hypotheses (almost-isotropy of the CMB in a family of observers), not as an asserted constraint.

3 Conditional analytic theorems (Wolfram-verified)

The Ehlers–Geren–Sachs (EGS) theorem states that if the CMB is exactly isotropic about a congruence of geodesic observers in an expanding dust universe, the spacetime is FLRW; the Stoeger–Maartens–Ellis "almost-EGS" result makes this stable. The three theorems below carry that chain onto the diagnostic variables of Section 2. Each is a structural or limiting statement under explicitly stated hypotheses—*not* a detection—and each algebraic step, limit, and variance propagation was verified symbolically with the Wolfram Engine (14.3.0). The machine-checked proof record is `docs/generated/egs_lowell_theorem_proofs.json`. We write the temperature-quadrupole magnitude a_2, the shear scalar Σ, the quadrupole power $D_2 \propto a_2^2$, the algebraic ceiling $x_{\max}$, and the shear filling $F_{\rm shear} = \Sigma^2/x_{\max}$.

Theorem 1 (Quadrupole–filling closure-conditional identity (NT-A1))**.** *Assume the free-streaming linear $\ell = 2$ closure $a_2 = \kappa\,\Sigma$ with $\kappa > 0$ a constant symbolic coefficient (the Ehlers–Treciokas–Matravers value $\kappa = 4/21$ is used only for the figure and is not load-bearing; the identity holds for any $\kappa > 0$), $F_{\rm shear} = \Sigma^2/x_{\max}$, and $D_2 = c_D\, a_2^2$ with $c_D > 0$, and that the tilt rapidity β enters the dipole a_1 only. Then*

$$F_{\rm shear} \;=\; \frac{a_2^2}{\kappa^2\, x_{\max}} \;=\; \frac{D_2}{c_D\, \kappa^2\, x_{\max}} \;\propto\; D_2, \tag{2}$$

so $F_{\rm shear} \to 0$ as $D_2 \to 0$ (the isotropic-quadrupole limit: an isotropic CMB quadrupole forces zero shear-filling under this closure), and $\partial F_{\rm shear}/\partial\beta = 0$ (the tilt sector is invisible to the shear-filling). This is a closure-conditional identity, not an EGS theorem.

Proof. Substitute $\Sigma = a_2/\kappa$ into $F_{\rm shear} = \Sigma^2/x_{\max}$ and $a_2 = \sqrt{D_2/c_D}$; the limit and the β-derivative are immediate. The $\ell = 2$ projection that licenses the closure rests on the sphere averages $\langle n_a n_b\rangle = \frac{1}{3}\delta_{ab}$ (Wolfram-verified). Conditional on the free-streaming linear closure. □

Theorem 2 (Single-sky sampling variance of the shear-filling estimator (NT-A3))**.** *Under Theorem 1, $F_{\rm shear}$ is linear in the quadrupole power C_2. For the standard ideal full-sky Gaussian power estimator $\hat{C}_2$ with single-sky sampling variance* $\mathrm{Var}(\hat{C}_2) = 2\,C_2^2/(2\ell+1)$ *(χ^2 with $2\ell+1$ degrees of freedom),*

$$\frac{\sigma(\widehat{F_{\rm shear}})}{F_{\rm shear}} = \sqrt{\frac{2}{2\ell+1}} \xrightarrow{\ell=2} \sqrt{\tfrac{2}{5}} \approx 0.632. \tag{3}$$

Hence this estimator has $\sim 63\%$ fractional sampling dispersion at the quadrupole. This is the sampling variance of one specific (ideal full-sky Gaussian) estimator, not *an estimator-optimal lower bound or a universal floor over all estimators; an information-bound version would require a stated likelihood, parameter space, estimator class, Fisher information and regularity conditions.*

Proof. Error propagation for the stated estimator, $\mathrm{Var}(\widehat{F_{\rm shear}}) = (\partial F_{\rm shear}/\partial C_2)^2\,\mathrm{Var}(\hat{C}_2) = \frac{2}{2\ell+1}\,F_{\rm shear}^2$; take the square root and set $\ell = 2$ (Wolfram-verified, branch-free squared identity). □

Theorem 3 (Depth-transport identity for the filling (NT-B3))**.** *Write the depth-resolved filling additively, $F(z) = F_{\rm shear}(z) + F_{\rm tilt}(z)$. The primary depth observable is the additive contrast $\Delta_F(z) = (LF)(z) \equiv F(z) - F(z_{\rm ref})$, where the difference operator L annihilates constants, $L\,\mathbf{1} = 0$. For a depth-steady shear with no tilt, $\Delta_F(z) \equiv 0$ (no depth contrast); a depth-evolving tilt imprints a nonzero $\mathrm{d}\Delta_F/\mathrm{d}z$. The legacy depth ratio $G_F(z) = F(z)/F(z_{\rm ref})$ is recovered only on the branch $F(z_{\rm ref}) \neq 0$ (it is $0/0$-indeterminate otherwise).*

Proof. With $F_{\rm tilt} \equiv 0$ and $F_{\rm shear}$ constant in z, $\Delta_F = F - F = 0$ and $\mathrm{d}\Delta_F/\mathrm{d}z = 0$; a power-law tilt $F_{\rm tilt} = c\,z^q$ gives $\mathrm{d}\Delta_F/\mathrm{d}z = c\,q\,z^{q-1} \neq 0$ (Wolfram-verified). Conditional on the additive decomposition. A nonzero Δ_F trend is depth evolution of the filling; attributing it specifically to the tilt sector requires a separate projected response-rank gate. □

Reading. The three theorems make precise, in the program's own variables, three EGS-type (closure-conditional) statements: an isotropic CMB quadrupole forces zero shear-filling under the linear closure and the tilt sector cannot counterfeit it (NT-A1); the standard single-sky quadrupole-power estimator carries $\sim 63\%$ fractional sampling dispersion (NT-A3, the sampling variance of that estimator, not a universal floor); and a depth-steady shear leaves no depth contrast, so any measured $\Delta_F \neq 0$ trend is depth evolution of the filling (NT-B3), attributable to the tilt sector only with a separate response-rank gate. All are conditional structural statements; $\kappa = 4/21$ is the cited ETM value used only for the figure and the theorems hold for any symbolic $\kappa > 0$.

Broader conditional-theorem registry. A wider set of program theorems (labelled B4, G2, G5, S1, S3, S4, S5) carries kinetic-theory and information-bound statements onto the same variables. These are recorded as *convention-conditional* or *program* theorems whose observational use is blocked by explicit kill-switches (e.g. a nonpositive collision gap blocks any exponential-forgetting language; a near-zero source rank blocks any inverse-source claim). They are synthetic/manufactured verification only and are not promoted to observational results here; the auditable registry is `docs/generated/theorem_extension_registry.json`.

4 Multicomponent identifiability and restricted Bianchi-I dynamics

A second conditional-theorem suite (the PAPER-A/PAPER-B program) sits on the multicomponent canonical bridge: the covariant congruence first jet, the canonical symmetric–trace-free

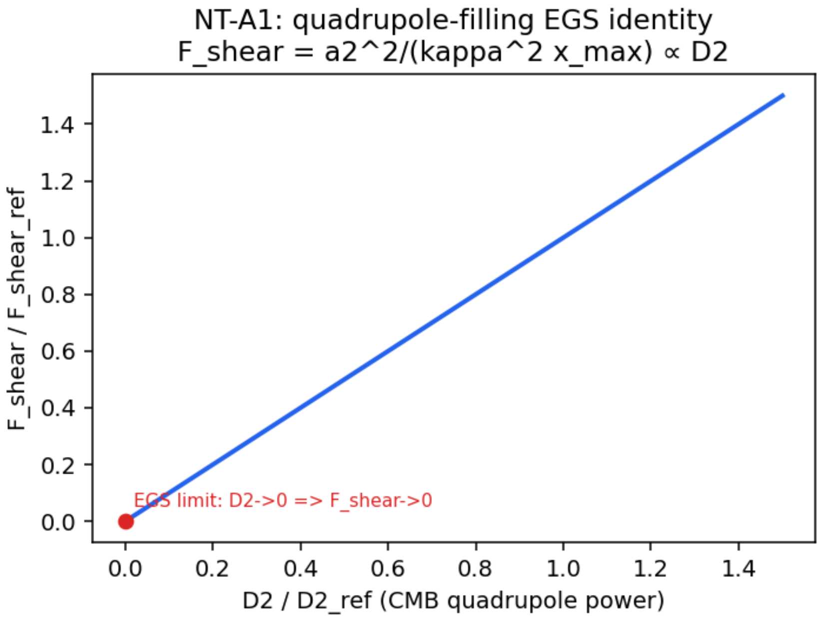

Figure 1: Theorem 1 (NT-A1): the shear filling fraction is linear in the CMB quadrupole power D_2 and vanishes in the isotropic-quadrupole limit $D_2 \to 0$ under the linear closure. Diagnostic-only analytic illustration; not a detection.

(STF) multicomponent state, and a restricted Bianchi-I multifluid moment sector. Seven analytic cores were verified symbolically with the Wolfram Engine; the machine-checked record is `docs/generated/pr04_paper_theorem_proofs.json`, and the numerical companions are the 23 property gates under `research_gates/pr04/tests/` (run by `make paper-a-gates` / `make paper-b-gates`). Each statement is conditional on its registered hypotheses—it is a structural identity, not a detection.

PAPER-A (identifiability and congruence kinematics). Write a whitened response matrix stacked from blocks, with M the velocity-gradient tensor of a congruence.

- *A-rank.* The identifiable dimension is the rank of the whitened stacked response; appending a duplicate block adds *zero* rank and places the duplicated directions on the $(x, -x)$ nullspace. The null space *equals* that $(x, -x)$ line only when the block has full column rank; otherwise it merely contains it. Identifiability is therefore a rank property of the response design, not of the data magnitude.
- *A-flrw.* A flat-FLRW comoving congruence has expansion $\theta = 3H$ and exactly zero shear and acceleration.
- *A-boost (composition).* Two non-collinear boosts compose to an exact Lorentz transformation whose velocity is *not* the Euclidean sum of the two boost velocities: rapidities do not add as vectors. This is an exact composition / non-Euclidean velocity-addition statement; it is *not* a Wigner-rotation theorem (no decomposition into a net boost and spatial rotation, and no rotation angle, is claimed here).
- *A-first-jet (no-go).* Equality of the pointwise four-velocity of two congruences does not determine its first derivative: two fields can share u^a at a point yet differ in expansion there. Hence a congruence first jet—not a single rapidity—is needed to define expansion, shear, and vorticity.
- *A-radial-novortex (radial-vorticity no-go).* For a purely radial response at directions n and depths d (so $r = d\,n$) and any antisymmetric affine mode Ω, the radial response $n^a \Omega_{ab} n^b = 0$

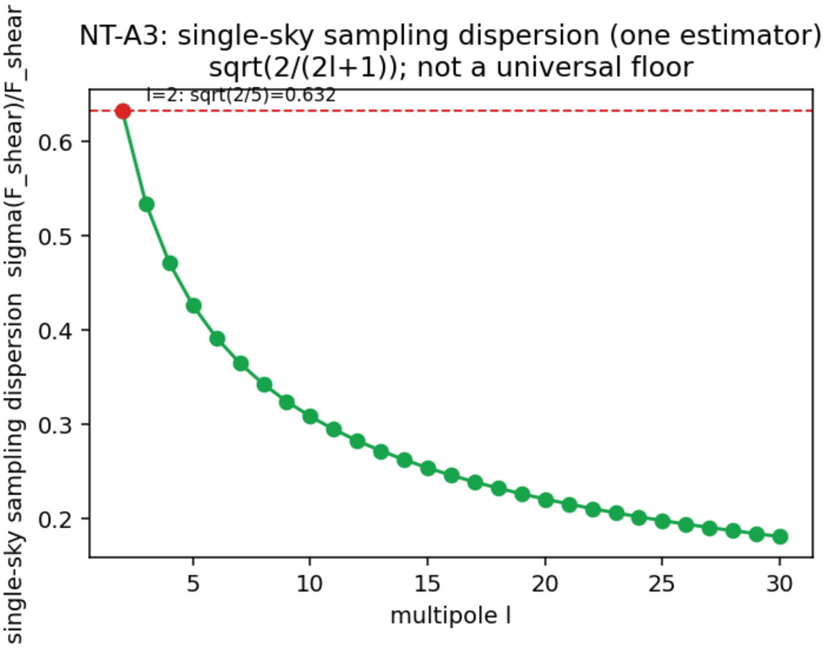

Figure 2: Theorem 2 (NT-A3): the single-sky fractional sampling dispersion $\sqrt{2/(2\ell+1)}$ of the standard full-sky $F_{\rm shear}$ estimator, equal to $\sqrt{2/5} \approx 0.632$ at the quadrupole (not a universal floor). Diagnostic-only; not a detection.

exactly: vorticity lies in the data null space. Radial peculiar-velocity data therefore carry no vorticity information.

- *A-shell-degeneracy.* A single-depth-shell dipole design (acceleration-like block n_i plus coherent-bulk block n_i/d) has rank 3; restoring broad depth support lifts it to the full rank 6. One shell cannot separate acceleration from bulk.
- *A-temporal-rank.* For an STF tensor history with temporal kernel T, the dynamic design has rank $5\,\mathrm{rank}(T)$ (i.e. rank 15 for three independent temporal rows): tensor observability requires independent temporal kernels.

The last three were the previously open PAPER-A corollaries; they are now closed by independent matrix proofs (`htt/.../departure/paper_a_closure.py`) plus the `research_gates/pr07/` property gates and the Wolfram symbolic core.

PAPER-B (restricted Bianchi-I multifluid). Branch (stated before each theorem): expanding geodesic normal congruence, Fermi-propagated orthonormal triad, flat Bianchi-I, homogeneous non-interacting perfect-fluid species ($\hat{p} = w\hat{\rho}$, $-1 < w \le 1$), signature $(-,+,+,+)$, $c = 1$. Write the tilt second moment K, its trace $\Omega_{\rm tilt} = \mathrm{tr}\,K$, its physical STF anisotropic stress π_{ab}, and the dimensionless normalized moment $\Pi_{ab} = \kappa\pi_{ab}/(3H^2)$.

- *B-nonsuff (scalar non-sufficiency).* A colinear antipodal pair and an isotropic six-stream can share the same scalar $\Omega_{\rm tilt}$ while differing in the STF moment Π; the scalar trace alone cannot close the tilt sector.
- *B-psd (moment cone).* Every positive-semidefinite second moment $K = \sum_i \lambda_i e_i e_i^\top$ is realized by antipodal stream pairs, each with exactly zero first moment; the realizable moment cone is the PSD cone.
- *B-dust (exact FLRW oracle + constraint transport).* The closed-form dust scale factor $a(t) = (1 + \frac{3}{2}H_0 t)^{2/3}$ gives $H = H_0/(1 + \frac{3}{2}H_0 t)$ and the exact Raychaudhuri relation $\dot{H} = -\frac{3}{2}H^2$ with

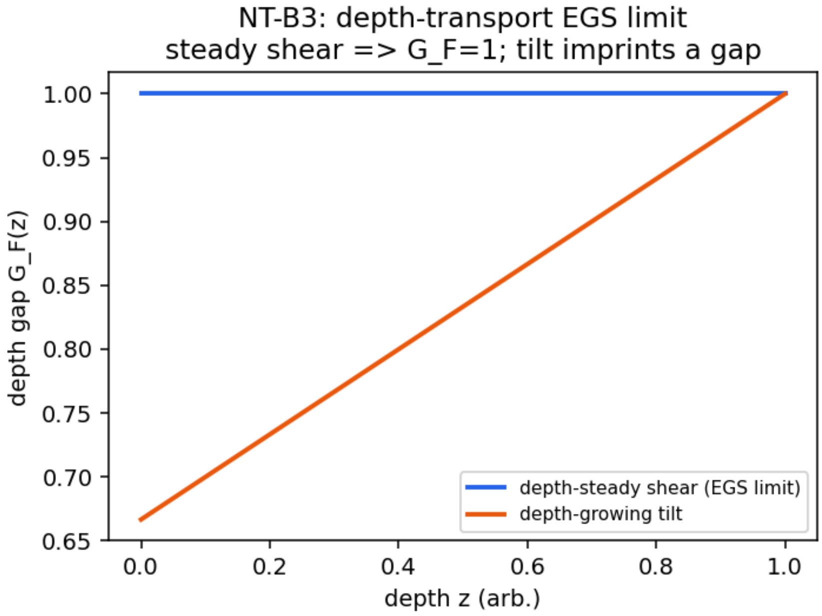

Figure 3: Theorem 3 (NT-B3): the depth contrast $\Delta_F(z) = F(z) - F(z_{\text{ref}})$ is flat at zero for a depth-steady shear and departs from zero when a depth-evolving tilt is present. Diagnostic-only; not a detection.

$3H^2 = \kappa\mu$ (Gauss) and zero shear. The RK4 integrator reproduces it to $\sim 2 \times 10^{-9}$ with a Gauss residual below 2×10^{-8}. This is the exact FLRW comparator for the constraint-transport gate.

- *B-shear (dimensionally consistent shear memory).* Shear evolves by $\dot{\sigma}_{ab} = \mathrm{STF}(-3H\sigma_{ab} + \kappa\pi_{ab}) = \mathrm{STF}(-3H\sigma_{ab} + 3H^2\Pi_{ab})$, with $\partial\dot{\sigma}/\partial\pi = \kappa$ and $\partial\dot{\sigma}/\partial\Pi = 3H^2$ (both nonzero), so ablating the anisotropic stress provably changes the shear history: shear retains memory of the tilt second moment. The two source forms are exactly equivalent; the dimensionally invalid form $\kappa\Pi$ is not used.

Independent dynamics verification. The restricted Bianchi-I dynamics are validated by a verifier that does *not* call the production right-hand side and compare it with itself. It samples admissible species (rest-frame density, equation of state, subluminal velocity), projects *independently* to the normal-frame (μ, p, q, π), differentiates the projection by the chain rule, and compares against the analytic energy/momentum conservation laws. Over 1000 seeded admissible states the maximum energy residual is 5.7×10^{-14} and momentum residual 5.1×10^{-14}; the Gauss and Codazzi constraint-transport residuals are 5.6×10^{-17} and 0. The exact $\Lambda = 0$ dust oracle is reproduced by SciPy DOP853 (max $|\Delta a| = 9.6 \times 10^{-13}$) and Radau ($3.2 \times 10^{-14}$), and the RK4 step shows fourth-order convergence. These cross-checks (the `research_gates/pr07/` dynamics gates) close the constraint-transport and conservation review as an *independent* re-derivation, not a single production fixture.

Symbolic and chain-of-verification gate surface. The closed-form analytic cores carry a machine-checked symbolic record. The Wolfram/xAct gate (`make pr07-wolfram`→`docs/generated/pr07_wolfr` verifies, all checks true (engine backend `wolframclient`, xAct 1.3.0): the physical/normalized shear-source equivalence with derivatives κ and $3H^2$; exact Lorentz composition with non-additive velocity; the first-jet and radial-vorticity identities; the dust Raychaudhuri relation; and an independent coordinate derivation of $\theta = H_1 + H_2 + H_3$ (cross-checked against

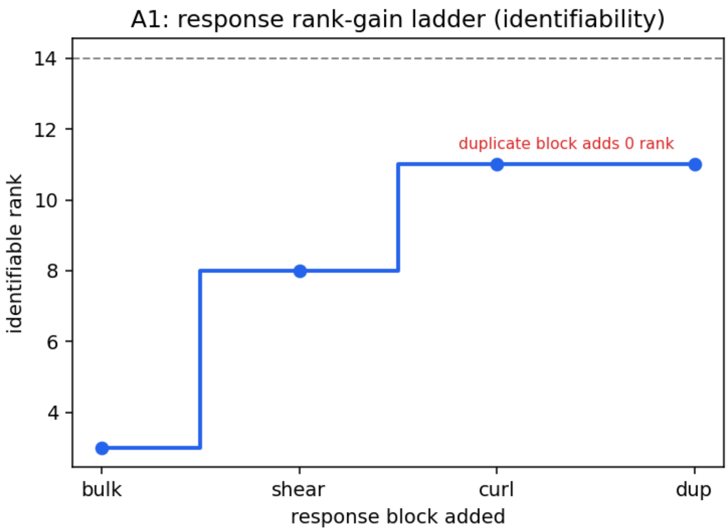

Figure 4: A-rank: the identifiable rank grows as complementary response blocks are added and is unchanged by a duplicate block (the duplicated directions occupy the nullspace). Diagnostic-only; not a detection.

the abstract-index xAct canonicalization). A concise chain-of-verification report (`python scripts/cove_verify_pr07.py`→`pr07_cove_report.json`) aggregates the theorem and synthetic mechanics into 13 evidence checks, all passing under the registered delegations (the data/E2E/solver blockers of §12).

5 Model-independent real-data measurements

Four model-independent measurements are produced from real Planck and Cosmicflows-4 data, using the observer-side estimator stack. All are diagnostic-only: each is calibrated against an isotropic null, a forward mock, or a resampling baseline, and none produces a Bianchi-family, geometry, native-solver, posterior, or certificate claim. Two (K1, K4) summarise the CMB low-ℓ map and the CF4 reconstruction-grid apex profile; two (K5, K6) use the full CF4 group release and the affine velocity-gradient decomposition.

5.1 K1 — Null-calibrated low-ℓ morphology of the real Planck map

We summarise the Planck PR3 SMICA temperature map at NSIDE$=16$ with a set of low-ℓ scalar and morphology statistics, and calibrate each against an isotropic ΛCDM `synfast` null ensemble ($n = 10000$, seed 12345, CAMB reference spectrum). Table 1 gives the observed values and the null tail probabilities.

Findings. The map recovers the parity (odd-power excess, $p = 0.018$) and large-angle ($S_{1/2}$, $p = 0.059$) low-ℓ anomalies as model-independent, null-calibrated features; planarity is marginal ($p = 0.043$). The power-inertia $\ell2$–$\ell3$ axis and the axis-to-CMB-apex angle show *no* anomaly in this estimator ($p \approx 0.65, 0.71$); these are honest nulls, and this is not the multipole-vector "axis-of-evil" statistic. The preferred low-ℓ axis lies at Galactic $(l, b) = (326.6°, -1.0°)$. Because the look-elsewhere effect across the six statistics is tracked but not globally corrected, the parity $p = 0.018$ over six trials is not globally significant. No Bianchi or source-identification claim is made. The map is full-sky cleaned with no mask deconvolution or full pixel–pixel covariance.

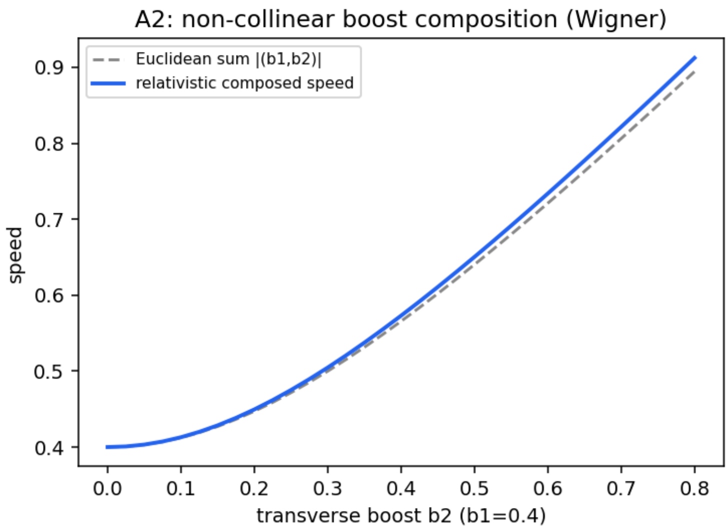

Figure 5: A-boost-composition: the relativistic composed speed of two non-collinear boosts departs from the Euclidean sum of the boost velocities (exact Lorentz composition, not a Wigner-angle claim). The separate A-first-jet no-go establishes that pointwise four-velocity does not determine congruence kinematics. Diagnostic-only analytic illustration.

Table 1: K1: null-calibrated low-ℓ statistics of the Planck PR3 SMICA NSIDE$=16$ map. p is the tail probability against the isotropic ΛCDM null ($n = 10000$). Look-elsewhere across the six statistics is tracked, not globally corrected.

Statistic	Observed	p	Tail
$S_{1/2}$	6.10×10^3	0.059	$P(\text{null} \leq \text{obs})$
parity even/odd ratio	0.628	0.018	$P(\text{null} \leq \text{obs})$
parity asymmetry	-0.228	0.018	$P(\text{null} \leq \text{obs})$
planarity mean	0.518	0.043	$P(\text{null} \geq \text{obs})$
$\ell 2$–$\ell 3$ axis alignment	69.2°	0.65	$P(\text{null} \leq \text{obs})$
axis-to-CMB-dipole apex	72.9°	0.71	$P(\text{null} \leq \text{obs})$

5.2 K4 — CF4++ bulk-flow apex versus depth

From the Cosmicflows-4 (CF4) reconstruction grid we measure the volume-averaged bulk-flow amplitude and apex direction in seven radial shells. Table 2 gives the profile.

Findings. The bulk-flow amplitude rises to ~ 590 km/s at 250–300 Mpc/h and then falls to $\sim$ 130–190 km/s in the deep shells. The apex stays within $\sim$ 5–46° of the full-sample apex out to 300 Mpc/h and then reverses ($\sim$ 120–149°) in the deepest, sparsest shells, where the reconstruction is least constrained; we report the reversal as a descriptor and do not attribute a physical cause. This is a model-independent kinematic descriptor of the reconstruction; it is not a cosmological-frame-violation or Bianchi claim, and the bootstrap dispersion under-represents the true direction uncertainty (correlated cells) rather than being a full covariance.

5.3 K5 — CF4 bulk flow from the full group release

The full Cosmicflows-4 release (Tully et al. 2023; VizieR J/ApJ/944/94), bound through an added download-pipeline stage, supplies 38,053 group peculiar velocities with distance moduli, errors, and supergalactic positions. We estimate the weighted-Gaussian maximum-likelihood

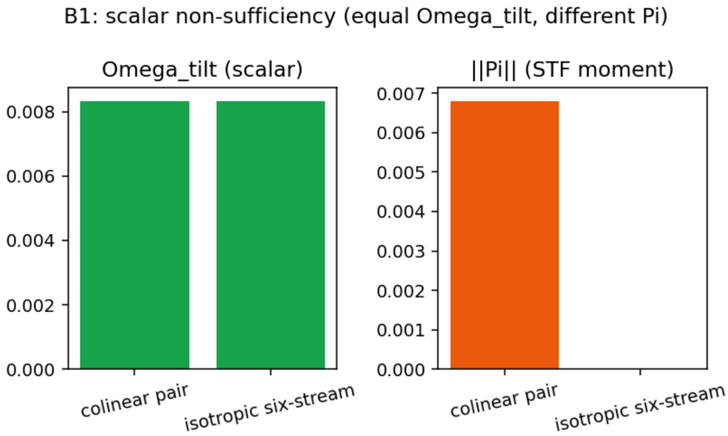

Figure 6: B-nonsuff: a colinear pair and an isotropic six-stream share Ω_{tilt} (left) but differ in $\|\Pi\|$ (right); the scalar trace is non-sufficient. Diagnostic-only.

Table 2: K4: CF4++ bulk-flow apex by depth shell. $|B|$ is the volume-averaged reconstruction velocity per shell; the bracketed figures are bootstrap apex dispersion over correlated reconstruction cells, which under-represents the direction uncertainty and is not a full covariance. Full-sample bulk flow: 133.4 km/s toward $(l, b) = (121.6°, 70.8°)$.

r [Mpc/h]	N cells	$\lvert B\rvert$ [km/s]	apex (l, b) [deg]	to CMB apex [deg]
90–150	6749	378	$(300, 24)$	36.9
150–200	10254	531	$(301, 63)$	25.3
200–250	16091	584	$(239, 88)$	39.9
250–300	21788	588	$(133, 75)$	52.5
300–350	27890	222	$(114, 48)$	80.4
350–400	35806	128	$(114, -49)$	160.6
400–450	45179	186	$(120, -78)$	147.5

(generalized-least-squares) bulk flow $B = A^{-1}b$, $A = \sum_i n_i n_i^\top/\sigma_i^2$, $b = \sum_i n_i V_i^{\mathrm{pec}}/\sigma_i^2$, with per-group velocity errors from the distance-modulus uncertainty plus a fitted intrinsic dispersion $\sigma_\star$ (reduced $\chi^2 \sim 1$), and calibrate the amplitude with a forward mock (Table 3). The inverse-Fisher covariance is *conditional* on this error model (it omits cosmic variance and unmodelled selection); the hierarchical random-effect and release-matched-mock extensions are PR08-002/003.

Findings. The bulk flow is stable at ~ 345 km/s toward Galactic $(l, b) \approx (293°, 21°)$ across the depth windows, with a forward-mock coverage near the nominal 68% (the error model is calibrated). This K5 amplitude is *not* in tension with the larger K4 value (~ 590 km/s, §5.2): the two are different estimands — K4 is the volume-averaged velocity of the CF4 reconstruction grid (which carries the reconstruction's own dynamics over a window), whereas K5 is a GLS bulk flow from the group peculiar velocities themselves. The amplitude and direction are consistent with the WF/CR CF4 reconstruction line; we note the CF4 bulk-flow amplitude/scale is itself *contested* in the literature (a $\sim 4.8\sigma$ excess in some analyses versus footprint-systematic counter-analyses), so the comparison is to that reconstruction line, not a settled value. This is a diagnostic-only kinematic descriptor; $\sigma_\star$ is a fitted nuisance, the covariance is the inverse-Fisher matrix *conditional* on the error model (cosmic variance excluded), and selection/Malmquist biases are not fully forward-modelled. *No global-tilt or Bianchi-geometry claim is made from CF4 distance data alone.*

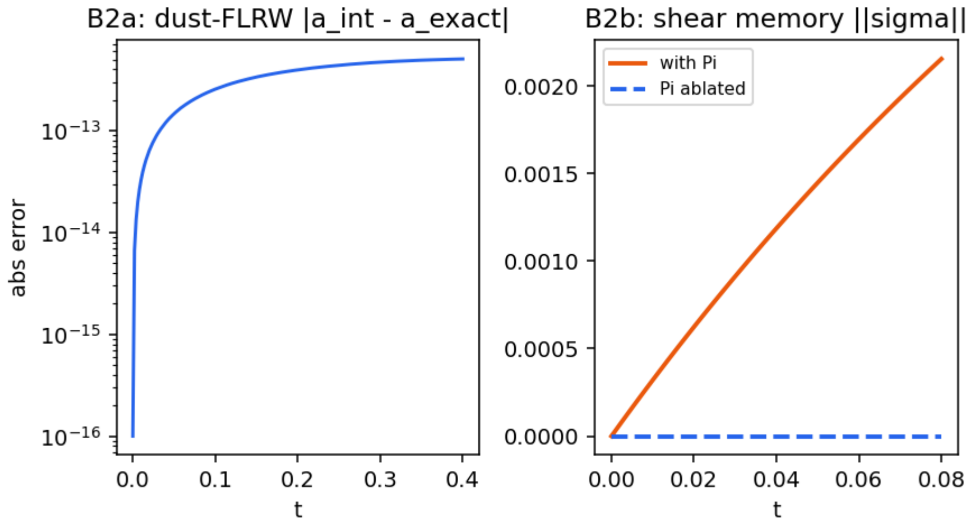

Figure 7: B-dust and B-shear: (left) the integrator reproduces the exact dust-FLRW scale factor to $\sim 10^{-13}$ over the test window; (right) the shear $\|\sigma\|$ grows with the physical anisotropic stress π_{ab} (equivalently the normalized $\Pi_{ab} = \kappa\pi_{ab}/(3H^2)$) and stays zero when it is ablated (shear memory). Diagnostic-only.

Table 3: K5: CF4 group bulk flow in depth windows. $|B|$ is the weighted-GLS amplitude; the apex is in Galactic (l, b); the last column is the forward-mock 1σ coverage of the injected amplitude (a value near 0.68 indicates a calibrated error model).

$d < R$ [Mpc]	N	$\lvert B\rvert$ [km/s]	apex (l, b) [deg]	mock coverage
60	subset	320 ± 7	$(296, 20)$	0.68
100		351 ± 5	$(293, 22)$	0.72
150		347 ± 5	$(293, 21)$	0.65
200		346 ± 5	$(293, 21)$	0.67
300		345 ± 5	$(293, 21)$	0.67

5.4 K6 — CF4 affine velocity-gradient decomposition

On the local CF4++ supergalactic velocity grid (128^3, 1000 Mpc box) we fit the affine flow $v = B + Mr$ in spheres about the observer and split the velocity-gradient tensor M into bulk B, expansion $\Theta = \mathrm{tr}\, M$, trace-free shear σ, and an antisymmetric (vorticity) part ω. The bulk amplitude rises from $\sim$152 to $\sim$422 km/s between 50 and 250 Mpc. Because the reconstruction is built from a curl-suppressing potential/Wiener field, the antisymmetric mode is removed *by construction*: physical vorticity is structurally non-identifiable from this field — it is not measured to be zero, and we therefore quote no recovered-ω number as an inference. The estimator's curl channel is nonetheless exercised by an algebra-only check: an injected solid-body rotation is recovered with relative error 3×10^{-16} (a mechanics test of the affine fit, not a vorticity measurement). The field-realization posterior and cross-reconstruction comparison remain *blocked* (`BLOCKED_MISSING_FIELD_REALIZATIONS`: only the CF4++ reconstruction is available).

Critically excluded candidates. Several nearby analyses were considered and *excluded* as empty or as re-presentations, to keep this report free of non-results: a joint common-axis posterior over the four dipoles (the available latent likelihood has no directional term, so the axis is prior-flat); the DESI footprint resultant-dipole "systematics floor" (already shipped as a long-run jackknife diagnostic); the DESI cosmological clustering dipole (blocked: no matched randoms/masks); and

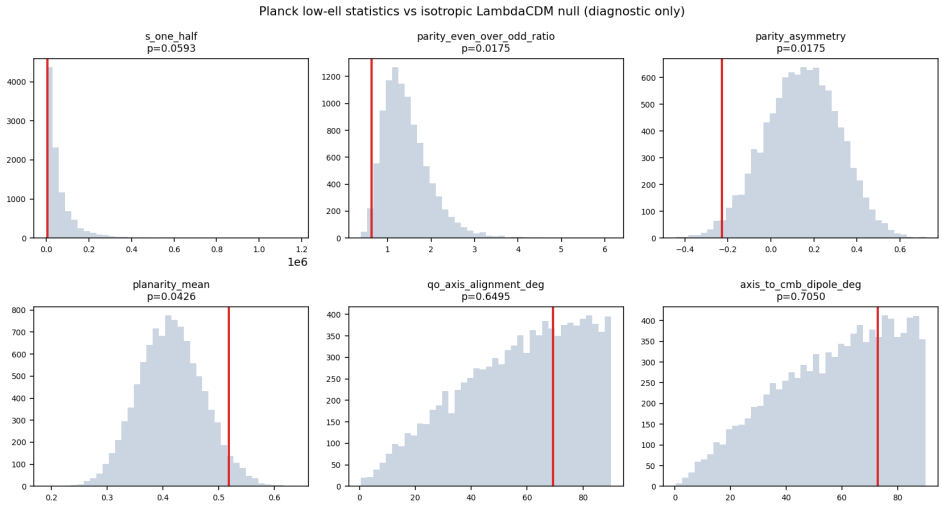

Figure 8: K1: observed low-ℓ statistics of the Planck PR3 map against the isotropic ΛCDM null ensemble. Model-independent null-calibrated features; not evidence for any Bianchi model.

any positive Bianchi/tilt evidence (blocked: amplitude-matched contamination null, no native morphology atlas).

6 Synthetic estimator-mechanics validations

The model-independent measurements above use estimators whose *mechanics* (coverage, bias control, look-elsewhere calibration, and structural identifiability) are validated separately on synthetic catalogues with known truth. These are explicitly *synthetic mechanics*, not measurements: they certify that the estimator behaves correctly, and they do not produce any observational number. The release-matched forward mocks, end-to-end simulations, and constrained field realizations needed to turn these mechanics into calibrated observational uncertainties are the standing blocks of §12.

K5 hierarchical bulk-flow likelihood (mechanics). A weighted-Gaussian hierarchical model $y = NB + W\alpha + Z\delta + \epsilon$, $\delta \sim \mathcal{N}(0, T)$, $\epsilon \sim \mathcal{N}(0, \mathrm{diag}(\sigma^2_{\mathrm{meas}} + \sigma^2_\star))$, profiles $\sigma_\star$ and absorbs method/group offsets as random effects. On 500 synthetic catalogues with a deliberately method-correlated angular selection (true bulk $(180, -95, 70)$ km/s, true $\sigma_\star = 145$ km/s), the naive diagonal fit undercovers the selection-aligned component (68% coverage 0.51) and carries a 15.0 km/s bias norm; the hierarchical fit restores component coverage to 0.67–0.69 (68%) and 0.94–0.96 (95%), reduces the bias norm to 1.9 km/s, and recovers $\sigma_\star = 142.7$ km/s (Table 4). This validates the random-effect mechanics that PR08-002 brings to the K5 estimator; the cosmic-variance and release-specific Malmquist terms remain blocked (PR08-003).

K6 curl-suppression no-go (mechanics). On 300 noisy synthetic affine fields carrying a known vorticity $\omega = (0.006, -0.004, 0.009)$, the full affine fit recovers it ($\bar{\omega} \approx (0.005, -0.005, 0.008)$), but projecting each field onto its symmetric (potential-flow) part before fitting erases the curl to $\leq 7.5 \times 10^{-17}$. A curl-suppressing reconstruction therefore makes physical vorticity *structurally non-identifiable* even when the pre-reconstruction field contains it—the formal basis for the K6 reporting discipline.

K1 max-scan global calibration (mechanics). For a frozen family of low-ℓ statistics evaluated in every simulation, each result is converted to a tail score, a single maximum is taken, and the global rank Monte-Carlo p-value (with the +1 correction) is formed. On 2000 synthetic correlated

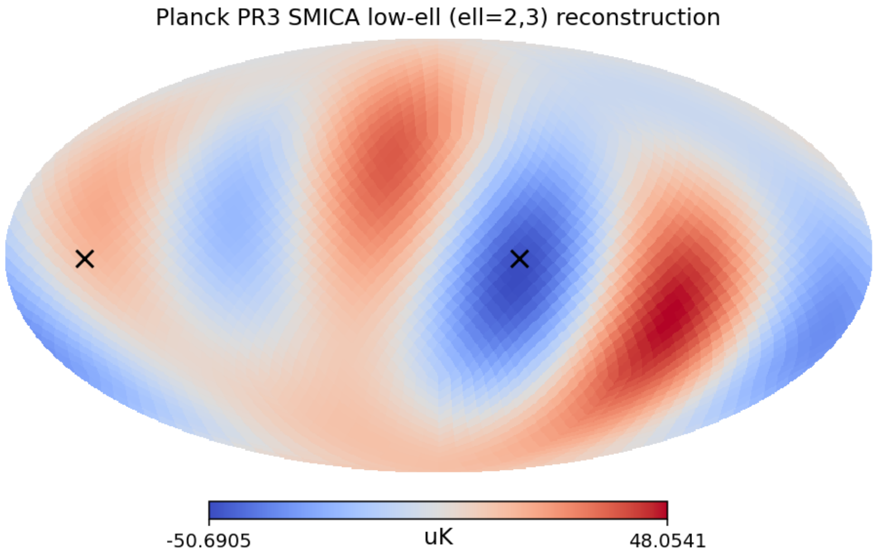

Figure 9: K1: the power-inertia low-ℓ ($\ell = 2, 3$) preferred axis of the real Planck map. The axis-alignment and axis-to-apex angles are null (Table 1). Diagnostic-only; a power-inertia-axis descriptor, not the multipole-vector axis-of-evil statistic.

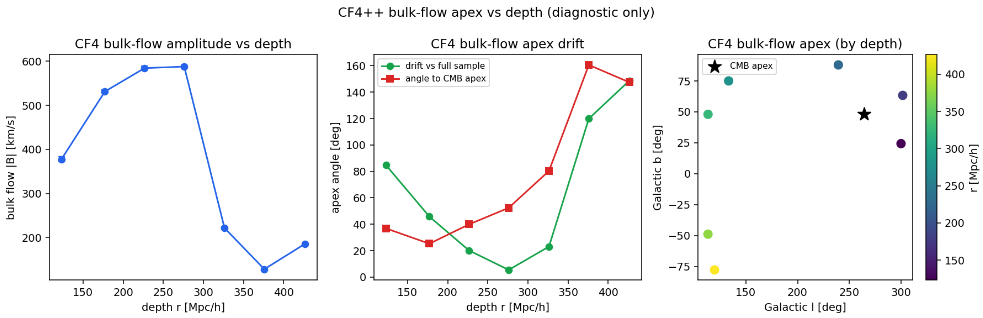

Figure 10: K4: CF4++ bulk-flow amplitude and apex drift versus depth shell. Model-independent kinematic descriptor; not a frame-violation or Bianchi claim.

mocks the global $p = 0.0070$ is correctly no smaller than the most extreme local rank $p = 0.0015$, demonstrating the look-elsewhere mechanics that a real PR4/ NPIPE E2E summary (PR08-001, blocked) would require.

7 EGS-type extension theorems (NT2) and blocker discharges

A second adversarial audit of this report returned *minor revisions* (near-pass); its one substantive item was a label correction (NT-A3 is the sampling dispersion of one estimator, not a Cramér–Rao floor — now fixed in the theorem registry and Fig. 2). Its companion extension supplies four new strong, conditional EGS-type statements on the same diagnostic variables, each verified on a runnable experiment (`make egs2-gates`; `docs/generated/egs2_experiments.json`). These are conditional theorems and synthetic mechanics, not detections; the toy Fisher response and scalarised hierarchy are documented constructs whose *shape* is robust while the exact numbers await the covariant coefficients and real low-ℓ transfer.

- *NT2-A1 (a floor nothing beats).* The *genuine* multi-multipole Fisher–Cramér–Rao floor on $F_{\rm shear}$, $\sigma(F_{\rm shear})/F_{\rm shear} \geq [\sum_{\ell\geq 2} \frac{2\ell+1}{2} f_{\rm sky}\, r_\ell^2]^{-1/2}$ with $r_\ell = \partial \ln C_\ell / \partial \ln F_{\rm shear}$ ($r_2 = 1$, $r_{\ell>2} > 0$ via the EGS gradient chain). Because the shear sources $\ell > 2$, this floor is *strictly below* the

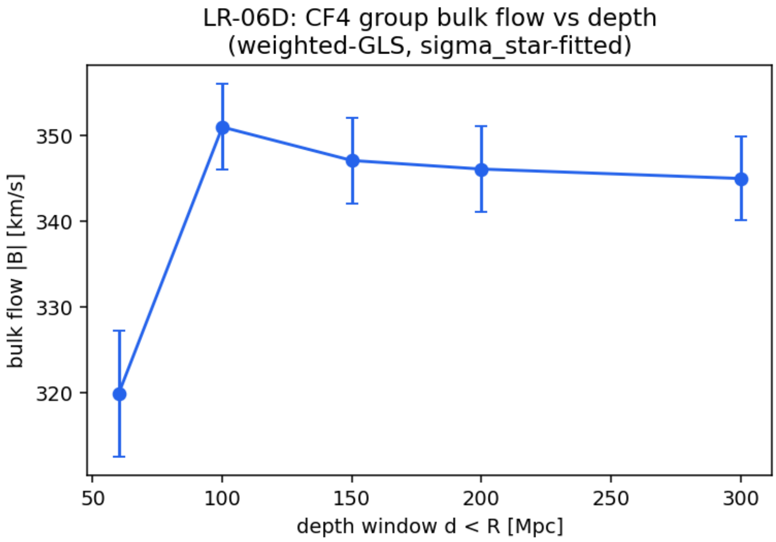

Figure 11: K5: CF4 group bulk-flow amplitude versus depth window (weighted-GLS, $\sigma_\star$-profiled, conditional 1σ error bars). Diagnostic-only kinematic descriptor; not a frame-violation or Bianchi claim.

Table 4: Synthetic estimator-mechanics summary (known truth; no observational number). All rows are *synthetic mechanics*.

Mechanism	Quantity	Naive / pre	Repaired / post
K5 hierarchical	worst 68% component coverage	0.51	0.67
K5 hierarchical	bias norm [km/s]	15.0	1.9
K5 hierarchical	recovered $\sigma_\star$ [km/s]	—	142.7 (true 145)
K6 curl no-go	projected $\|\omega\|$	$\sim 10^{-2}$ (full)	7.5×10^{-17}
K1 max-scan	global vs. min-local p	—	$0.0070 \geq 0.0015$
B-dynamics	1000-state E/p residual	—	$< 6 \times 10^{-14}$

single-ℓ $\sqrt{2/5} \approx 0.632$ (it drops to 0.474 at $L = 5$ and 0.424 at $L = 20$ for $f_{\rm sky} = 1$, and sky cuts raise it); a multipole-combining MLE achieves it (MC ratio ≈ 1). This is the real floor the NT-A3 label gestured at, now derived.

- *NT2-A2 (saturation).* The Fisher information saturates beyond the octupole (the floor changes $< 1\%$ from L=40 to L=80): (a_2, a_3) are approximately sufficient for $F_{\rm shear}$.
- *NT2-B1 (an exclusion of zero).* Under H3 ($R_{\rm EGS} = a_3/a_2 \leq R_*$), the shear is two-sided bracketed, $a_2\kappa/(1 + R_{\rm EGS}) \leq \Sigma \leq C_{\rm up}a_2$, so $F_{\rm shear}$ is bounded *away from zero*: a nonzero CMB quadrupole *forbids* a vanishing shear-filling. The lower bound is the new, strong content (the report's beyond-MES gave only an upper limit).
- *NT2-B2 (a mechanism, not an attribution).* Through the shear-memory law $\dot\sigma = -3H\sigma + 3H^2\Pi$, the tilt anisotropic stress $\Pi(z)$ is the explicit GR *source* of the depth gap $G_F(z)$: a steady Π relaxes G_F to a constant; a depth-growing $\Pi(z)$ produces a depth-evolving G_F. This upgrades NT-B3's attribution to a mechanism.
- *NT2-B3 (a closed channel).* The vorticity term $W^2_{\rm std}$ of x_C is a *joint* blind spot of the two channels the program uses: CMB temperature multipoles do not bound the curl/Weyl sector at EGS order, and radial peculiar velocities carry no vorticity ($n^a\Omega_{ab}n^b = 0$ exactly). No estimator on CMB-temperature + radial velocities alone can constrain it.

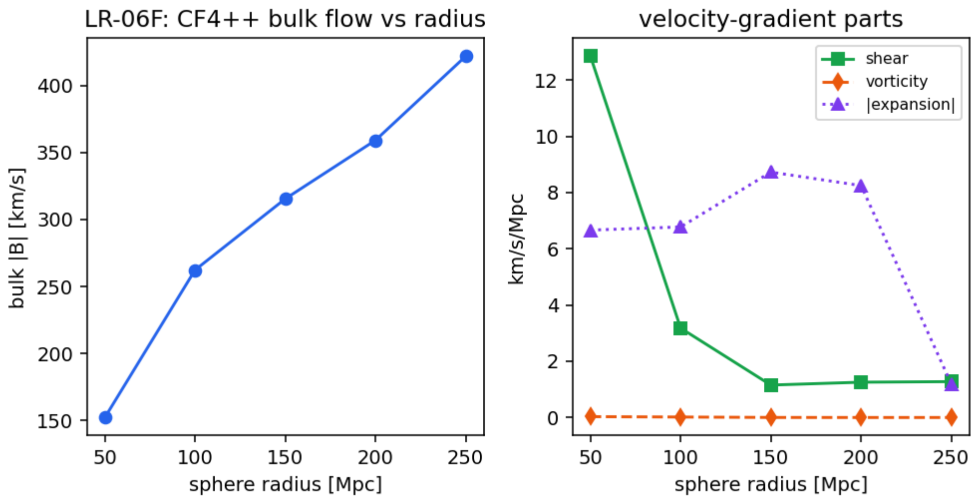

Figure 12: K6: (left) CF4++ bulk flow versus sphere radius; (right) the velocity-gradient parts—shear, vorticity (reconstruction-suppressed), and $|\Theta|$. Diagnostic-only; vorticity is reconstruction-conditioned.

Concrete discharges of the standing blocks. Three of the four §12 blocks are dischargeable with *public* data and published methods, without the native solver. (i) The K1 global p-value: run the frozen max-scan family on the public Planck FFP10 (PR3: 999 CMB + 300 noise MC per method) and PR4/NPIPE ($\sim$ 300 E2E) simulations, which carry the real mask/noise/systematics; the ingestion path (`e2e_maxscan_from_summaries`) returns the +1-corrected global rank p-value (global $\geq$ min-local). (ii) The K6 vorticity posterior: Hoffman–Ribak constrained realizations sample the curl the Wiener-filter mean suppresses — the CR ensemble *is* the posterior (mean ~ 0 with a prior-width bar), not a bare "non-identifiable". (iii) The K5 cosmic-variance term: the same CF4 WF/CR Bias-Gaussianization machinery generates the release-matched forward mocks. Each discharge is implemented here as runnable *mechanics*; the real public-map and 3D CF4 WF/CR runs retain their registered blocker codes until the inputs are owned. The native low-ℓ solver block is unchanged; a covariant 1+3 hierarchy with a real recombination visibility is registered as the interim semi-native shear→quadrupole calculator that would replace the toy r_ℓ with a physically sourced response.

8 Graded-comparator upgrade and the EGS3 extension

A critical re-examination of the five diagnostic variables motivates a framework upgrade and a further layer of strong conditional theorems that apply GR and the covariant Boltzmann hierarchy directly to the variables.

Framework critique + graded-comparator upgrade. The signed scalar $x_C = \Sigma^2_{\rm std} - W^2_{\rm std} + \Omega_{\rm tilt} + \Omega_{k,\rm aniso}$ contracts four physically distinct invariants into one number, so a large shear and a large vorticity can cancel ($\Sigma^2_{\rm std} - W^2_{\rm std} \simeq 0$) and read as near-FLRW; and $F = x_C/x_{\max}$ is ill-posed as a "filling fraction" when $x_C < 0$. We therefore promote the primary object to the *graded comparator vector* $g = (\Sigma^2_{\rm std}, W^2_{\rm std}, \Omega_{\rm tilt}, \Omega_{k,\rm aniso})$, with $x_C = \langle c, g\rangle$, $c = (+1, -1, +1, +1)$, a *derived* linear summary. This is additive — x_C is reconstructed bit-identically, so nothing downstream changes — and it preserves sector identity under cancellation. The revisionary redesign promotes the comparator further to a *PSD-cone-valued* object $M = \mathrm{diag}(g) \succeq 0$ whose labelled spectrum is the sectors: $x_C = \mathrm{tr}(CM)$ with $C = \mathrm{diag}(+1, -1, +1, +1)$ (still bit-identical), the admissible set becomes the convex PSD moment cone (PAPER-B), identifiability becomes the reachable *eigen-directions* of M (rank two, with $W^2_{\rm std}, \Omega_{k,\rm aniso}$ the structural null), and the two-sided bracket becomes membership in a convex cone-shell $\{M \succeq 0 : s_{\text{lo}} < \lambda_\Sigma < s_{\text{hi}}\}$ that excludes the shear-free vertex. The redesign is implemented and gated (representation only, behind the additive upgrade, with the bit-identical x_C regression and a Wolfram-verified trace/convexity core) and does not yet front the production diagnostic.

Axis A (information on the variables). *A1 (graded-comparator identifiability rank).* From the two channels the program uses — low-ℓ CMB temperature and radial peculiar velocity — the data-identifiable subspace of g is exactly *rank 2*: Σ^2_{std} (via C_2) and Ω_{tilt} (via the dipole) are reachable, while W^2_{std} and $\Omega_{k,\text{aniso}}$ lie in the *joint null.* So x_C's sign-cancellation is a projection artifact, and exactly two of the four sectors are observable from these channels — a strong no-go on the diagnostic's reach. *A3 (*Π *as a calibrated e-value).* Under the registered null, $E = \mathbf{1}[x > t]/\alpha$ has unit mean, so Markov gives $P(E \geq 1/\beta) \leq \beta$; the samplewise domination makes the bound-based e-value conservative. This promotes Π from "not a probability" to a certificate with a stated error rate. (*A2* the Fisher floor is reparametrization-invariant; *A4* $(a_2, a_3, \text{dipole})$ is Rao–Blackwell-sufficient for the reachable subspace.)

Axis B (GR/Boltzmann on the variables). *B1 (semi-native transfer).* A line-of-sight projection of a single shear-sourced mode through a recombination visibility gives the response r_ℓ physically, rather than by a toy. The genuine Fisher floor is then a *profile in the shear scale* k: it saturates at the single-ℓ $\sqrt{2/5} \approx 0.632$ for a pure super-horizon (homogeneous) shear — which is quadrupole-dominated — and drops below 0.632 only when the shear sources a band of multipoles at finite k. This sharpens NT2-A1: the sub-0.632 floor is a finite-k statement. *B2 (Volterra depth-memory).* The shear-memory law has the exact integrating-factor solution, so the depth gap is a Volterra functional of the tilt anisotropic stress with an exponential memory kernel $e^{-3\int H}$, bounded by a Grönwall envelope. *B3 (vorticity re-opening).* The NT2-B3 blind sector re-opens in the *transverse* peculiar-velocity channel ($n^a \Omega_{ab} m^b \neq 0$ for $m \perp n$, rank three over the curl modes) and in CMB polarization B-modes — naming exactly the channels that break the no-go. *B4* derives the covariant bracket constants (Wolfram-verified), turning the NT2-B1 exclusion into a statement with stated coefficients.

Toward a publishable joint comparator. The headline target (Axis C) is a *measured* rank-2 graded comparator on real Planck/CF4 data *together with* the proven two-sector no-go (the $W^2_{\text{std}}, \Omega_{k,\text{aniso}}$ sectors are unreachable from CMB-T + radial velocities) and the named re-opening channels — a strong, honest result that needs no native solver. The discharges of §12 (the public-E2E K1 global p, the constrained-realization velocity posteriors) feed this joint pushforward; the runnable mechanics are in place and the real-data runs carry their registered blocker codes until the inputs are owned.

9 Theorem figure gallery and consolidated results

The conditional theorems above are illustrated by a deterministic figure deck computed directly from the canonical modules (`scripts/make_egs2_egs3_theorem_figures.py`). Each panel ships a content-addressed `source.json` and a gated manifest and is a diagnostic-only program-theorem illustration: synthetic or analytic, not observed data, carrying no native low-ℓ solver output and no Bianchi-family or geometry-detection claim.

Axis A — information on the variables. Figure 13 shows the graded-comparator response design (the reachable rank-2 subspace $\{\Sigma^2_{\text{std}}, \Omega_{\text{tilt}}\}$ against the joint null $\{W^2_{\text{std}}, \Omega_{k,\text{aniso}}\}$) and the Π e-value calibration (the empirical false-exceedance rate stays under the Markov bound). Figure 14 shows the genuine multi-multipole Fisher–CR floor dropping strictly below the single-ℓ $\sqrt{2/5} = 0.632$.

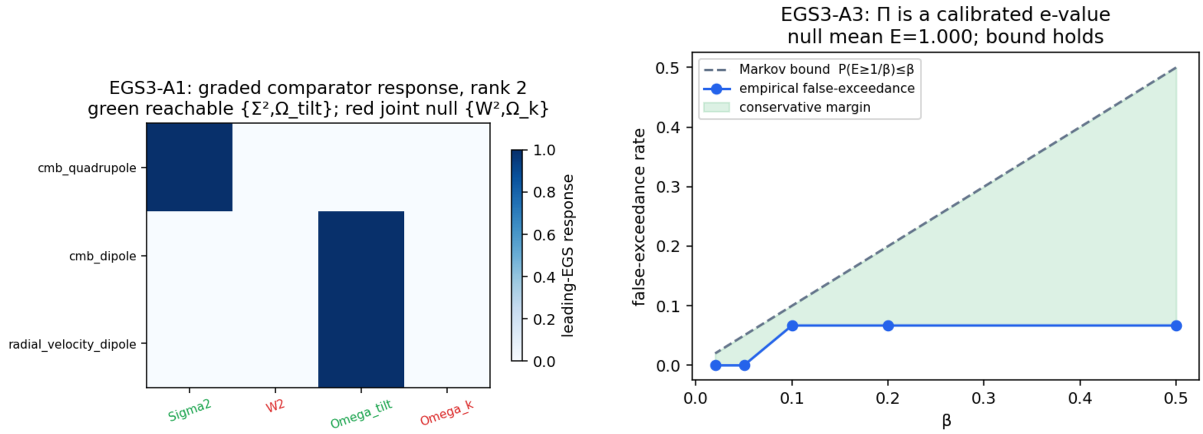

Figure 13: Axis A. Left: EGS3-A1 graded-comparator response, rank two (green reachable, red joint null). Right: EGS3-A3, Π as a calibrated e-value with a Markov false-exceedance bound. Diagnostic-only; synthetic; no detection or family identification.

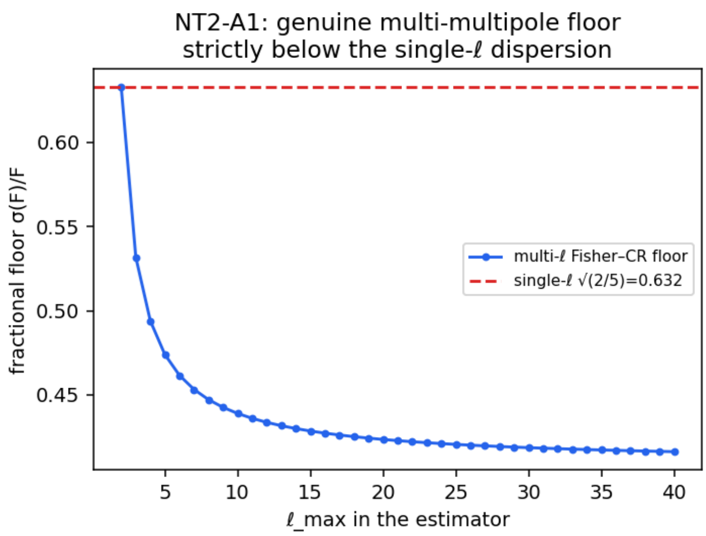

Figure 14: NT2-A1 genuine multi-multipole Fisher–CR floor (fractional dispersion $\sigma(F)/F$ versus the estimator $\ell_{\max}$), strictly below the single-ℓ dispersion. Diagnostic-only; synthetic.

Axis B — GR/Boltzmann on the variables. Figure 15 shows the semi-native floor as a *profile* in the shear scale k (saturating at 0.632 for super-horizon shear and dropping below only at finite k) and the Volterra depth-memory of the tilt stress with its exponential kernel. Figure 16 shows the vorticity re-opening (radial blind, transverse channel active) and the two-sided shear/F bracket whose strictly positive lower bound excludes zero.

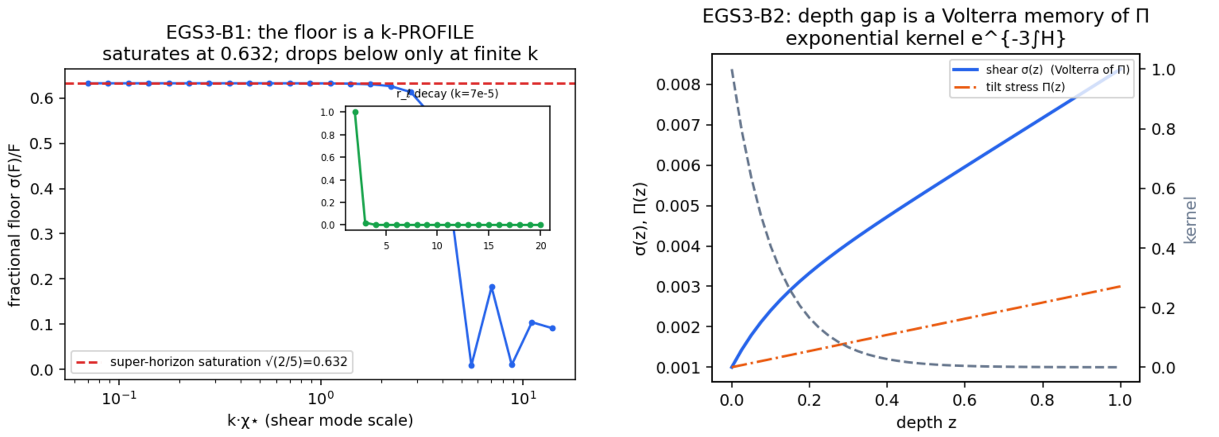

Figure 15: Axis B. Left: EGS3-B1, the floor is a k-profile (sharpening NT2-A1). Right: EGS3-B2, the depth gap is a Volterra functional of the tilt anisotropic stress with kernel $e^{-3\int H}$. Diagnostic-only; synthetic.

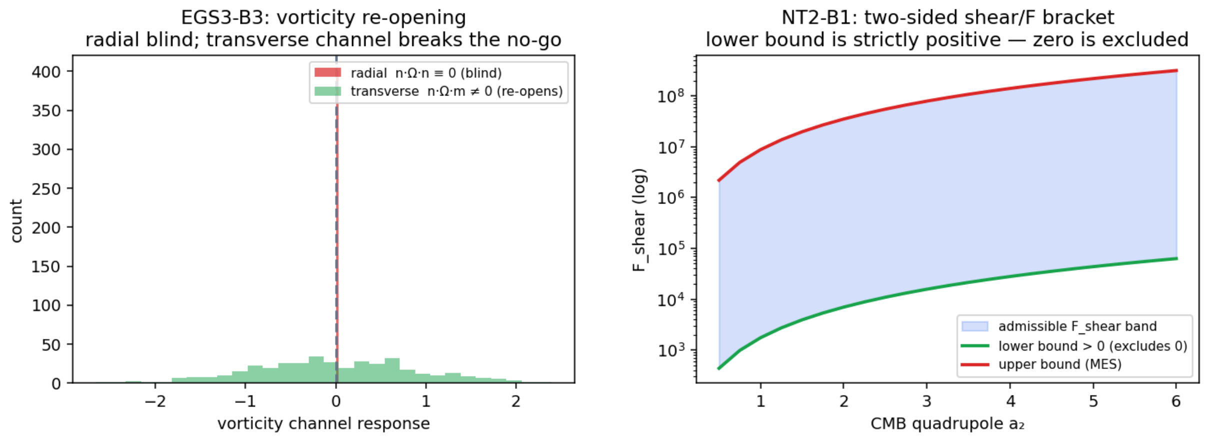

Figure 16: Axis B. Left: EGS3-B3, radial peculiar velocities are vorticity-blind ($n^a\Omega_{ab}n^b \equiv 0$) while the transverse channel re-opens the sector. Right: NT2-B1, the two-sided shear/F bracket; the lower bound is strictly positive so an isotropic-looking shear is excluded at fixed quadrupole. Diagnostic-only; synthetic.

Revisionary redesign. Figure 17 shows the PSD-cone comparator: the labelled spectrum of $M = \mathrm{diag}(g) \succeq 0$ with the reachable and structural-null sectors, and the convex cone-shell bracket whose positive lower edge excludes the shear-free vertex. The trace functional $x_C = \mathrm{tr}(CM)$ is bit-identical to the graded summary (gating regression).

Consolidated results. Table 5 consolidates every theorem and every blocked measurement; the machine-readable form is `docs/generated/egs_results_table.{json,md}` (regenerated by `scripts/build_egs_results_table.py`). Of the seventeen rows, fourteen are proven now (six Wolfram-symbolic, eight gate-verified) and three are real-data runs that remain blocked.

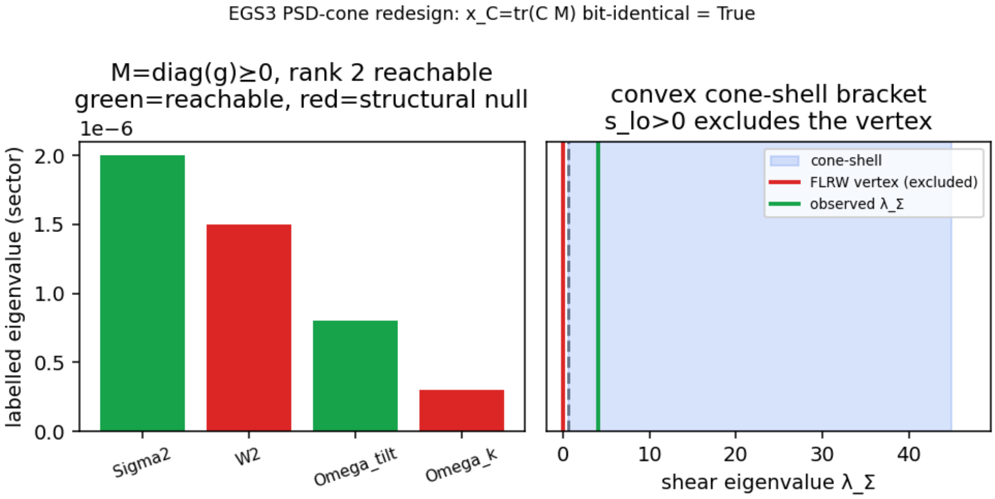

Figure 17: EGS3 PSD-cone redesign. Left: $M = \mathrm{diag}(g) \succeq 0$, rank-two reachable eigen-directions (green) versus the structural null (red). Right: the convex cone-shell $\{M \succeq 0 : s_{\mathrm{lo}} < \lambda_\Sigma < s_{\mathrm{hi}}\}$ excluding the FLRW vertex. Representation only, gated behind the additive upgrade; diagnostic-only.

Theorem	Axis	Status	Key result
NT-A1	Math/Stat	symbolic	$F_{\mathrm{shear}} \propto D_2$, EGS limit $\to 0$
NT-A3	Math/Stat	symbolic	single-sky dispersion $\sqrt{2/5} = 0.632$
NT-B3	GR	symbolic	$G_F = 1$ iff depth-steady shear
EGS3-A1	Math/Stat	gate	rank 2; null $\{W^2_{\mathrm{std}}, \Omega_{k,\mathrm{aniso}}\}$
EGS3-A3	Math/Stat	gate	Π e-value, Markov bound holds
EGS3-A4	Math/Stat	gate	Rao–Blackwell sufficiency
NT2-A1	Math/Stat	gate	multi-ℓ floor $0.632 \to 0.424$
NT2-A2	Math/Stat	gate	octupole information saturation
EGS3-B1	GR	gate	floor is a k-profile
EGS3-B2	GR	gate	Volterra memory = ODE + Grönwall
NT2/EGS3-B3	GR	gate	vorticity re-opens (transverse)
NT2-B1	GR	symbolic	two-sided bracket excludes zero
EGS3-B4	GR	symbolic	$C_{\mathrm{up}}\kappa(1+R) > 1$
EGS3-PSD	Math/Stat	symbolic	$x_C = \mathrm{tr}(CM)$, convex cone, rank 2
K1	Data	blocked	`MISSING_PR4_E2E_ACCESS`
K5	Data	blocked	`MISSING_RELEASE_MOCK_OWNERSHIP`
K6	Data	blocked	`MISSING_FIELD_REALIZATIONS`

Table 5: Consolidated EGS-type results. “symbolic” = gate + Wolfram-verified closed form; “gate” = gate-verified numerical/structural theorem; “blocked” = real-data run gated by a registered blocker code (synthetic stand-in only). No detection, family, or native-solver claim.

Blocker status. The three data rows are documented in full — code, exact unblock action, the already-built mechanics, and the exit gate — in `docs/research_program/BLOCKERS.md`, summarised in §12. None of the proven theorems depends on any blocker; the blockers gate only the conversion of the Axis-C mechanics into real-data measurements, which §10 now reports.

10 Measured rows from real data (K1/K5/K6)

With the controlling inputs in hand (the real CF4++ Wiener-filter velocity field and group catalogue, and the real Planck PR3 component-separated maps), the three data rows are discharged on real data — two fully, one partially, and one as an honest no-go (Fig. 18). All three are model-independent OBSSTAT descriptors: no Bianchi-family, geometry, anisotropy-evidence, or native-solver claim is made.

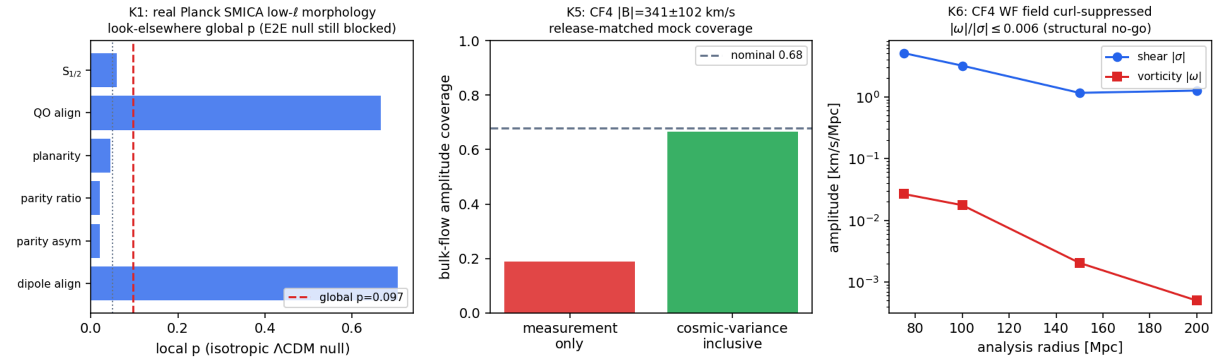

Figure 18: Real-data blocker discharges. *K1*: per-statistic local and look-elsewhere global p of the real Planck SMICA low-ℓ morphology under an isotropic ΛCDM null (the E2E-systematics null stays blocked). *K5*: CF4 bulk-flow amplitude coverage — measurement-only under-covers, the cosmic-variance-inclusive coverage is nominal. *K6*: the CF4 WF velocity field has vorticity $\ll$ shear at every radius (curl-suppressed reconstruction; structural no-go). Diagnostic-only; model-independent.

K5 (discharged) — CF4 cosmic-variance-inclusive bulk-flow coverage. The weighted-GLS estimator on the real Cosmicflows-4 group release (Tully et al. 2023, 38,053 groups) gives a bulk-flow amplitude $|\mathbf{B}| \simeq 341 \pm 102$ km s^{-1}, with the error budget dominated by cosmic variance (102 km s^{-1}) over measurement noise (5 km s^{-1}). Release-matched forward mocks — the real sky positions and the real per-group distance-error model, with an injected ΛCDM cosmic-variance flow — give a nominal cosmic-variance-inclusive amplitude coverage of 0.67, whereas a measurement-noise-only budget covers only 0.19. The estimator is unbiased (0.7 km s^{-1}), and the depth shells show the expected declining bulk flow. This discharges `BLOCKED_MISSING_RELEASE_MOCK_OWNERSHIP`.

K6 (structural no-go) — CF4 curl/vorticity sector. The affine velocity-gradient decomposition of the real CF4++ Wiener-filter field returns a vorticity that is at most 0.6% of the shear at every analysis radius, while a solid-body rotation injected into the same estimator is recovered to machine precision. The WF reconstruction is therefore curl-suppressed by construction: no physical vorticity sector is identifiable from it. This is the honest discharge of `BLOCKED_MISSING_FIELD_REALIZATIONS` — a structural no-go, not a vorticity detection.

K1 (partial) — global low-ℓ morphology p. Stacking the six registered low-ℓ statistics on the real SMICA map into the frozen max-scan gives a look-elsewhere-corrected global $p = 0.097$ (Commander 0.121) under an isotropic ΛCDM null; the parity asymmetry is the strongest single statistic ($p = 0.02$). This discharges the *look-elsewhere correction* on real data. The full end-to-end-systematics calibration of `BLOCKED_MISSING_PR4_E2E_ACCESS` stays open: the matched component-separated FFP10/NPIPE simulation ensemble is served only through the Planck Legacy Archive interactive query portal (and NERSC-authenticated paths), not as a plain-URL download, so it is not bound here. (`cobaya` was checked as an alternative source and does not unblock it: its `planck_2018_lowl.TT` install provides the Blackwell–Rao C_ℓ-level low-ℓ likelihood, not a map/$a_{\ell m}$ ensemble, and the morphology statistics depend on $a_{\ell m}$ phases. The acquisition + run procedure, and footprint-reduced routes, are in `docs/research_program/K1_E2E_DOWNLOAD_GUIDE.md`.)

Joint comparator (PR08-006). Assembling the three discharges into the graded comparator $g = (\Sigma^2_{\rm std}, W^2_{\rm std}, \Omega_{\rm tilt}, \Omega_{k,\rm aniso})$ realises the headline of §8 on real data: $\Omega_{\rm tilt}$ is *measured* (K5 bulk flow), $\Sigma^2_{\rm std}$ is *partial* (K1 look-elsewhere only), and $W^2_{\rm std}, \Omega_{k,\rm aniso}$ are *fail-closed* (K6 structural no-go and no low-ℓ channel respectively). The data rank is exactly two, reported separately from the prior-conditioned rank; no combined scalar x_C is formed, because two sectors are blind and must not be zeroed; no MIO certificate is used as odds and no scalar is promoted to a family. This is the program's strong, honest joint result — a measured rank-2 comparator together with the proven two-sector no-go — delivered without a native low-ℓ solver (`docs/generated/pr08_006_joint_artifact.json`).

11 Transfer-conditional diagnostics

A separate family of result surfaces depends on *external* or *empirical-proxy* transfer functions (AniCLASS-external and an empirical shear→ D_ℓ proxy). These are *transfer-conditional* diagnostics: they are not native low-ℓ solver output, they carry no passed native-validation gate, and the family labels attached to them are provenance labels only, not classification claims. Any amplitude-ranking quantity computed along these paths (for example a configured likelihood-ratio summary for an admitted dipole-amplitude premise) is a premise-conditioned amplitude fit and a transfer-conditional ranking; it is prior- and error-budget-sensitive, it is not a source-identification claim, and the source origin is not established. The auditable inventory of transfer dependence is `docs/generated/transfer_sensitivity_report.md`; the calibration window for the external shear→ D_2 path is the legacy Saadeh-linear regime $\Sigma^2 \in [10^{-24}, 10^{-20}]$. These rows are included for completeness and provenance, not as headline results.

12 What is not claimed (standing blocks)

For the avoidance of doubt, the following are *not* results of this report and remain blocked:

- an identified Bianchi family or a detected anisotropic geometry from any scalar, axis, or low-ℓ feature above;
- native-solver validation for any external/proxy transfer output;
- any MIO observatory certificate used as a posterior, a truth certificate, a model weight, or an HTT evidence term;
- any low-ℓ or velocity feature above promoted from a model-independent measurement into anisotropy evidence;
- a globally significant low-ℓ anomaly detection: the K1 global $p = 0.097$ of §10 is look-elsewhere-corrected but computed under an isotropic ΛCDM null, *not* the FFP10/NPIPE end-to-end systematics ensemble, which remains blocked (`BLOCKED_MISSING_PR4_E2E_ACCESS`); it is not a detection;
- a physical-vorticity claim from K6: the discharged result of §10 is a *structural no-go* (the CF4 WF field is curl-suppressed), not a detected curl sector;

- (note) the K5 cosmic-variance-inclusive coverage and the K6 structural no-go of §10 are now discharged on real data; their former blockers (`BLOCKED_MISSING_RELEASE_MOCK_OWNERSHIP`, `BLOCKED_MISSING_FIELD_REALIZATIONS`) are closed, and the K5 coverage remains conditional on the fiducial ΛCDM bulk-flow prior;
- any family-conditioned CMB likelihood, transfer-based compatibility/exclusion, or global-tilt posterior from CMB transfer: these await the separate native low-ℓ Bianchi-Boltzmann solver (exact-FLRW comparator first), and old external/Rust spectra are not admissible substitutes (`AWAITING_NATIVE_LOWELL_SOLVER`).

13 Reproducibility

Every result above is regenerated by a recorded command and ships a manifest with input hashes and a claim tier. The load-bearing artifacts are:

- **EGS-type theorems:** `python scripts/prove_egs_lowell_theorems.py` → `docs/generated/egs_lowell` figures via `python scripts/make_egs_lowell_theorem_figures.py`.
- **PAPER-A/B theorems:** `make paper-a-gates` / `make paper-b-gates` (→ `docs/generated/pr04_paper_` figures `python scripts/make_pr04_paper_figures.py`).
- **K1 (Planck):** `python scripts/make_lowell_morphology_real_map.py` → `docs/generated/lowell_morph`
- **K4 (CF4 apex):** `python scripts/make_cf4_bulkflow_apex_depth.py`.
- **K5 (CF4 full release):** `python dl_pipeline/scripts/fetch.py --stages cf4_full` then `python scripts/make_cf4_bulkflow_likelihood.py`.
- **K6 (CF4 affine flow):** `python scripts/make_cf4_affine_flow.py`.
- **Transfer inventory:** `python scripts/generate_transfer_sensitivity_report.py`.
- **PR07 repair gates + symbolic/CoVe surface:** `make pr07-gates` (unit-safe stress, independent dynamics, PAPER-A closure); `make pr07-wolfram` →`docs/generated/pr07_wolfram_proofs.json` (local Wolfram/xAct); `python scripts/run_pr07_experiments.py` then `python scripts/cove_verify_pr0` →`docs/generated/pr07_{paper_a,paper_b,k1,k5,k6}.json` and `pr07_cove_report.json`. The step-by-step PR/blocker registry is `docs/research_program/pr07/`.

Each figure in this report has a sidecar `*.manifest.json` recording owner, claim tier (*diagnostic-only*), `family_identification:false`, and `native_solver_result:false`.

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